Shengdong Liu

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Google Scholar

Education

University of Wisconsin-Madison Madison, Wisconsin, US
Ph.D., Environmental Chemistry and Technology Sep 2022 - Expected Dec 2026
- GPA: 3.929/4.0 Research focus: Optical sensors for pesticide detection.

Beijing Normal University Beijing, China
M.S., Environmental Science Sep 2019 - Jun 2022
- Research focus: Microplastics transport and fate in aquatic environments.

Chongqing University Chongqing, China
B.S., Environmental Science Sep 2015 - Jun 2019
- Research focus: Air pollution characterization, $PM_{2.5}$.

Selected Publications

1. **Liu, S.**, & Wei, H. (2025). Enhanced ligand exchange for ultrasensitive neonicotinoid detection in soil water via surface-enhanced Raman spectroscopy. *Water Research*, 286, 124268.
2. Wan, S., Chae, H., Meese, A. F., Nwokonkwo, O., Arrazolo, L., Yip, K. L., Ma, X., **Liu, S.**, Muhich, C., Wang, D., Wei, H., & Kim, J. H. (2025). Overcoming the reactivity-stability challenge in water treatment catalyst through spatial confinement. *Nature Communications*, 16 (1), 9672
3. **Liu, S.**, Lazarcik, J., & Wei, H. (2024). Emerging investigator series: quantitative insights into the relationship between the concentrations and SERS intensities of neonicotinoids in water. *Environmental Science: Nano*, 11(8), 3294-3300.
4. Ho, D., **Liu, S.**, Wei, H., & Karthikeyan, K.G. (2024). The glowing potential of Nile red for microplastics Identification: Science and mechanism of fluorescence staining. *Microchemical Journal*, 197, 109708.
5. **Liu, S.**, Shang, E., Liu, J., Bolan, N.S., Kirkham, M.B., & Li, Y. (2022). What have we known so far for fluorescence staining and quantification of microplastics: A tutorial review. *Frontiers of Environmental Science & Engineering*, 16(1), 8.
6. Li, Y., Liu, Y., **Liu, S.**, Zhang, L., Shao, H., & Zhang, W. (2022). Leaching mechanisms of bisphenol S from polyethersulfone (PES) and polyphenylsulfone (PPSU) microplastics during photoaging processes. *Environmental Science & Technology*, 56(5), 3033-3044.

Project Experience

U.S. EPA Science to Achieve Results (STAR) Program
Jun 2023 - April 2025

- Led the development of optical sensors for ultrasensitive neonicotinoid detection using surface-enhanced Raman spectroscopy (SERS).
- Collaborated with University of Florida (biochar preconcentration) and Yale University (advanced oxidation processes) to integrate sensing with treatment for neonicotinoid removal in rural watersheds.

Teaching & Mentoring

- **CEE 325 Environmental Engineering Materials, Lab Session, Teaching Assistant**
  - Instructed and assisted 50 undergraduate students in a lab setting.
  - Provided detailed instrument instructions and hands-on lab support.
  - Supported undergraduates' career development by serving as a professional reference for lab-position applications.
- **DELTA Mentoring Program, Undergraduates Mentor**
  - Completed 5-week comprehensive mentor training and earned a certified mentoring badge.
  - Mentored four undergraduate researchers through independent supervision, providing project design guidance, hands-on supervision, and fostering independent research skills.

Awards and Honors

- Outstanding Graduate of Chongqing University — 2019
- Excellent Undergraduate Thesis Award — 2019
- First-Class Graduate Scholarship, Beijing Normal University — 2021
- Environmental Science: Nano Outstanding Paper (2024; awarded 2025) — 2025
- Graduate School's Student Research Grants Competition Award — 2025

Skills

Instrumentation: Raman spectroscopy, High-Performance Liquid Chromatography (HPLC), Scanning Electron Microscopy (SEM), Dynamic Light Scattering (DLS).

Programming Language: Python.

Academic Services

Community Outreach: Provided scientific insights on neonicotinoid detection in the Black Earth Creek watershed to a community partner.